How can National CoE for Skilling bridge India's skill gap? Critically analyse the structural and technological challenges they face?

The Union Budget 2025-26 announced the set up of 5 National CoE for skilling with the aim of "Make for India, Make for the World".

It will bridge India's ~~skll~~ skill gap as—

→ Will implement industry linked skilling program

→ In technical fields, workforce would be trained.

→ Will reduce technological unemployment.

Currently it faces following structural and technological challenges—

① Industry academia gap → AICTE report 2022 showed that 60% engineering graduates are not suitable to work for industry.

(II) Lack of trained educator

(III) Unskilled workforce are not suitable for international standardisation.

WEF report showed that till 2030, there would be around 8.5 crore jobs available in digital sector. If India properly implement this policy, then it could ~~create~~ ~~become~~ create largest skilled workforce and would protect India from becoming skill deficit economy.

Can Atal Tinkering Labs transform STEM education in India? Analyse their potential impact on scientific temperament and innovation.

~~As~~ NITI Aayog report defined tinkering as learning through practical experiences. It's like hands on activities. Another report of NITI Aayog i.e., Tinkering in schools report 2023 mentioned that the logical reasoning and problem solving skills has been increased by 30% in students who ~~were~~ involved in tinkering labs. To promote this, government in budget 2025-26 decided to build 50,000 ATL in government schools.

Inclusion of AI, Machine learning, IOT, Robotics could promote scientific temperament and innovation. This could show potential impact as—

(I) Logical reasoning and critical thinking would enhanced.

(II) Students would enrich their skill capability.

(III) Next generation would be innovators and technologists.

(IV) Industry academia gap would get filled.

~~(V) Digital economy wou~~

Thus, ATL would promote cognitive ability and will build India's own innovation ecosystem.

The integration of AI in India's education system has the potential to revolutionise learning. Critically assess the feasibility of AI-driven educational reforms in India, highlighting ethical, infrastructural and pedagogical concerns.

Artificial Intelligence (AI) is a technology that enable systems to automatically learn from experiences and answer the queries like a human.

AI has the capability to revolutionise different sectors like agriculture, healthcare, banking systems, education sectors. The traditional and rote learning of education system could get undermined by this Artificial Intelligence.

AI has the potential to revolutionise the Indian education system in following ways!

① <u>Adaptive learning</u> → AI can provide personalised education to a student.

(II) Teacher Training Models → Through AI, teachers could enhance the skill capability

(III) Intelligent Tutoring System (ITS) → It would provide 24X7 help to students and would help in filling the learning gap.

(IV) Student performance using AI → It would help tutors to identify the patterns in student learning and adjust curriculum.

(V) Evaluation using AI → AI can provide immediate grading and feedback to students

(VI) Content creation → AI can automatically generate learning materials like practice questions, quizzes, etc.

This AI empowered education would provide many positive outcomes. These are as follows:

(I) Learning efficiency → Standford university released AI index report 2023 which mentioned that AI could increase learning efficiency by 25%.

(II) Skill development → As per UNESCO report, 2023 AI could increase cognitive and analytical skill of students to a certain level.

(II) Efficient use of teacher time → Teacher enabled with AI could provide enriched content to students within lesser ~~that~~ time.

With these educational reforms, AI could also bring some challenges regarding ethical, infrastructural and pedagogical matters too.

~~(I) Data Privacy → Privacy of users could be breached by using AI facility through provided by private companies.~~

CONCERNS	REASON	MEASURE
(I) Data Privacy	Privacy breach is a major concern in new technologies.	AI ethics education policy should be introduced.
(II) Accessibility to AI.	Due to digital divide, many students would not be able to avail its benefits.	Internet penetration should be reached upto rural level.
(III) Resistance to teacher for AI adoption	Role of teacher would get diminished and may led to unemployment.	→ Need to implement AI assisted teaching model → AI should work as supporting tool not as teacher replacement tool.

The target of NEP 2020 for skill based learning ~~throu~~ to promote job oriented education could get achieved through AI. By mitigating the ethical, infrastructural and pedagogical concerns and by optimally utilising the demographic dividend India has potential to ~~could~~ become a global digital learning leader.

Discuss the role of education and skill development in sustaining India's longterm economic growth. How can strategic investments in human capital bridge the gap between economic aspirations and realities?

United Nations Population Fund (UNFPA) report 2019 showed that India has 62.5% of its population in the age of 15-59 years. This huge demographic dividend needs proper education and skill development in sustaining India's longterm economic growth. Role of education and skill can be discussed as—

(I) It will promote innovation and scientific temperament.

(II) Migration of unskilled ~~work~~ workforce would reduced.

(III) Huge population would be secured from ~~beco~~ turning into demographic disaster.

(IV) Contribution of secondary sector in India's economy would get increased. Thus, it would ~~(V)~~ hinder jobless growth due to tertiary sector.

(V) Skill based learning would protect economy in becoming skill deficit economy.

(VI) ~~Education~~ Modern education like AI, Machine Learning, Robotics, cloud computing, etc would tackle ~~to~~ technological unemployment.

Union Budget 2025-26 showed the investment in human capital to achieve the economic aspirations. These are as follows:-

(I) ATL→ 50,000 ATL would be established in government schools for making students learn with activities and ~~pra~~ practical experiences. The inclusion of AI would reduce the industry academia gap.

As per NITI Aayog, Tinkering in schools report 2023, logical reasoning and problem solving skills of students has been increased by 30% who involved in tinkering labs. It would also promote digital economy of India.

(11) National Centre of Excellence for Skilling

This would enhance the skill capability of workforce. As per ~~AICCE~~ AICTE, 2022. report, 60% of engineering graduates are not suitable for industry. Investment in skilling would thus decrease this technological unemployment and would promote skill ~~enhanced~~ empowered economy. ~~A~~

As per WEF report, 8.5 crore jobs would be generated in digital and automation sector. Thus, ~~increase~~ investment in AI, ML, Cloud computing, Robotics would promote digital economy.

(III) Investment in IITs

Institutional funding in IITs would promote global research and will integrate industry with academia.

(IV) Centre of Excellence (CoE) in AI.

AI index report 2023 by Standford University showed that the learning efficeincy of students enabled with AI could get enhanced by 30%. Thus, it would promote technologically advanced generation and would preserve Indian ~~ecomy~~ economy from skill deficeit economy.

Thus, ~~if India~~ with regulatory framework and with certain international collaboration, India could optimally utilise its demographic dividend. The secondary sector would be benefitted and India would have skill empowered economy.